LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	21
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Descend forces in a descend glide range.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Descend forces in a descend glide range.
5. Specific Objectives	6.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of glide performance charts; Safety: Ensure no loose papers; Supervision: Instructor monitors chart reading.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four primary forces acting on an aircraft in flight?
(b) How does the relationship between lift and weight change during a descent?
(c) Define the term 'glide ratio' in the context of aircraft performance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Forces acting during a glide • Weight component along flight path • Lift and drag in unpowered flight • Relationship between glide angle and speed • Effect of weight on glide distance	L & Ds	PPT, IB	15 min
(b)	• Calculation of descent rate • Impact of configuration on glide • Best glide speed determination • Energy management in descent • Comparison of powered vs unpowered descent	L & Ds	PPT, PDF	20 min
(c)	• Factors affecting glide performance • Wind effects on glide range • Recovery techniques from glide • Emergency glide procedures • Limitations of glide capability	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the weight of the aircraft influences the glide angle.
(b) Describe the procedure for determining the best glide speed for maximum range.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of descent forces discussed in this lesson apply to the turn forces covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Turn forces in a turn, rate of turn steady level coordinated turn and control in a turn.	Refer to the master syllabus reference list	LP 22

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min